Eugene Ruiz Classes
CRC Cards & Class Details
Ver.1.2

CRC Cards

Class Name	Responsibilities	Collaborators
Flight	• Store flight data (number, origin, destination, terminals, times, status, aircraft) • Provide access to all attributes via getters/setters • Return available seats • Update status • Return formatted flight details	**Airport, Terminal, Schedule, Aircraft, Booking**
Schedule	• Store schedule data (ID, departure/arrival times) • Provide access to all attributes • Manage list of flights (add/remove) • Return schedule info	**Flight**
Booking	• Store booking data (ID, date, status, tickets) • Provide access to all attributes • Add/cancel tickets • Return booking status	**Ticket, Payment, Passenger**
Ticket	• Store ticket data (ID, flight, seat, status, passenger) • Provide access to all attributes • Assign/cancel seat/ticket • Return formatted ticket details • Provide passenger reference (Group 2 support)	**Flight, Seat, Passenger, Booking, TicketManager**
Payment	• Store payment data (ID, booking, amount, date, method, status) • Provide access to all attributes • Process payment • Update status and check completion	**Booking**
BookingManager	• Maintain collection of all bookings • Create, remove, find bookings • Get bookings for a specific passenger	**Booking, Passenger**
TicketManager	• Maintain collection of all tickets • Create, remove, find tickets • Provide getAllTickets() and getTicketsForPassenger() (Group 2 support)	**Ticket, Passenger**
Airline	• Hold references to all managers (light coordinator only) • Provide access to managers • Return airline name	**AirportManager, StaffManager, BookingManager, TicketManager**

Class Details

Flight

Instance Variables:
flightNumber: String Unique identifier for the flight (e.g., "AA123")
origin: Airport Departure airport for this flight
destination: Airport Arrival airport for this flight
departureTerminal: Terminal Terminal at the origin airport
arrivalTerminal: Terminal Terminal at the destination airport
departureTime: Schedule Schedule object containing departure timing
arrivalTime: Schedule Schedule object containing arrival timing
status: String Current flight status (e.g., "ON_TIME", "DELAYED", "CANCELLED")
aircraft: Aircraft Aircraft assigned to this flight

Methods:
Flight(String flightNumber, Airport origin, Airport destination, Terminal departureTerminal, Terminal arrivalTerminal, Schedule departureTime, Schedule arrivalTime, String status, Aircraft aircraft) • Full constructor that initializes all flight data • Ensures the object is always in a valid state
getFlightNumber(): String • Returns the unique flight number
setFlightNumber(String flightNumber): void • Updates the flight number
getOrigin(): Airport • Returns the departure airport
setOrigin(Airport origin): void

• Updates the departure airport
getDestination(): Airport • Returns the arrival airport
setDestination(Airport destination): void • Updates the arrival airport
getDepartureTerminal(): Terminal • Returns the departure terminal
setDepartureTerminal(Terminal departureTerminal): void • Updates the departure terminal
getArrivalTerminal(): Terminal • Returns the arrival terminal
setArrivalTerminal(Terminal arrivalTerminal): void • Updates the arrival terminal
getDepartureTime(): Schedule • Returns the departure schedule
setDepartureTime(Schedule departureTime): void • Updates the departure schedule
getArrivalTime(): Schedule • Returns the arrival schedule
setArrivalTime(Schedule arrivalTime): void • Updates the arrival schedule
getStatus(): String • Returns the current flight status
setStatus(String status): void • Updates the flight status
getAircraft(): Aircraft • Returns the assigned aircraft
setAircraft(Aircraft aircraft): void • Updates the assigned aircraft
getAvailableSeats(): int • Returns the number of available seats on this flight
updateStatus(String newStatus): void • Updates the flight status with a new value
getFlightDetails(): String • Returns a formatted string with key flight information